Prediction of DW-Nominate Scores

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Abstract— **This paper presents a study on predicting the ideological positions of U.S. Congressional members using DW-NOMINATE scores derived from Twitter data. DW-NOMINATE, a widely utilized method in political science, maps legislators' ideological stances across two dimensions: the first dimension captures the liberal-conservative spectrum, and the second highlights other ideological differences such as regional divides. Utilizing a dataset comprising 469,740 tweets from Congressional politicians spanning the period from 2008 to 2020, we aim to predict these two-dimensional ideological scores based on tweet metadata (e.g., favorite count, retweet count, hashtags) and textual content.**

I. INTRODUCTION

In recent years, social media has become a critical platform for political communication, with members of the U.S. Congress using Twitter to express their views and engage with the public. Traditionally, political ideology has been quantified using DW-NOMINATE scores, which are based on roll-call voting data.[1] These scores measure ideological positions across two dimensions: the first representing the liberal-conservative divide, and the second capturing regional and historical issues. While effective, this method relies on past voting behavior and doesn't account for real-time shifts in political discourse.

Given the widespread use of Twitter, it is worth exploring whether tweet content and engagement metrics can predict DW-NOMINATE scores. This study uses a dataset of 469,740 tweets from U.S. Congressional members between 2008 and 2020 to predict their ideological positions on both dimensions. We aim to determine if social media features such as tweet text, favorite counts, retweets, and hashtags can provide insights into lawmakers' ideologies.

The study has two main goals: (1) to predict the two dimensions of DW-NOMINATE using machine learning models, and (2) to conduct a descriptive analysis of how Congressional members use Twitter and how this correlates with their ideological stances. By combining natural language processing and data analysis techniques, this research bridges the gap between traditional political science and modern data-driven approaches.

II. DATA

The dataset used in this study consists of tweets from U.S. Congressional politicians, spanning a period from 2008 to 2020, amounting to a total of 469,740 tweets. This dataset provides rich textual and engagement information, offering a window into the political communication and ideological positions of lawmakers over time. The tweets were collected from publicly available Twitter accounts of members of Congress and are divided into three distinct datasets for training, testing, and submission.

A. Dataset Overview

The dataset is divided into three parts:

- **Training data:** Comprising 333,987 tweets, this subset is used to train the predictive models.
- **Test data:** This includes 135,753 tweets and is used to evaluate the performance of the models.
- **Sample submission data:** A smaller version of the test set that contains only the tweet Id and the political ideology dimensions for submission purposes.

This breakdown allows for effective training, testing, and evaluation of the models while preserving a structure that facilitates submission.

B. Features

Each tweet in the dataset is annotated with a series of features that capture both the content of the tweet and its audience interaction:

- **Id:** Unique identifier for each tweet.
- **favorite_count:** Number of times the tweet was favorited.
- **full_text:** The full content of the tweet, representing its text-based message.
- **hashtags:** List of hashtags included in the tweet.
- **retweet_count:** Number of times the tweet was retweeted.
- **year:** Year in which the tweet was posted.

These features collectively offer both quantitative and qualitative data points, which are critical for understanding both the message of the tweet and its impact on the audience.

C. Outcome Variables

Two primary outcome variables serve as the basis for this prediction task:

- **dim1_nominate:** The first dimension of the DW-NOMINATE score, reflecting the general liberal-conservative spectrum.
- **dim2_nominate:** The second dimension, which captures regional and historical variations in political ideologies.

D. Data Exploration and Visualization

During the height of the COVID-19 pandemic, Twitter became a battleground for public discourse, with hashtags capturing the pulse of the nation's concerns. Among these, #COVID19 dominated the conversation, reflecting the

	tweet_length_chars	tweet_length_words	hashtag_length_chars	hashtag_length_words
min	4.000000	1.000000	1.000000	1.00000
mean	180.578714	24.885511	14.536123	1.49198
median	151.000000	21.000000	12.000000	1.00000
max	2434.000000	61.000000	184.000000	17.00000

Fig. 1. Summary Statistics for Tweet and Hashtag Lengths (Characters and Words)

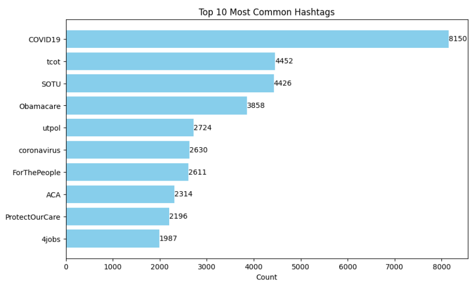

Fig. 2. Top 10 Most Commonly used Hashtags(hashtag column)

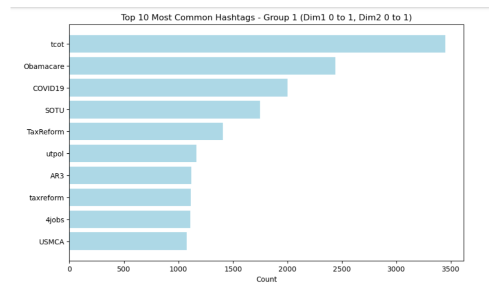

Fig. 3. Top 10 Most Common Hashtags - Group 1 (Dim 1 (0 to 1), Dim 2 (0 to 1)

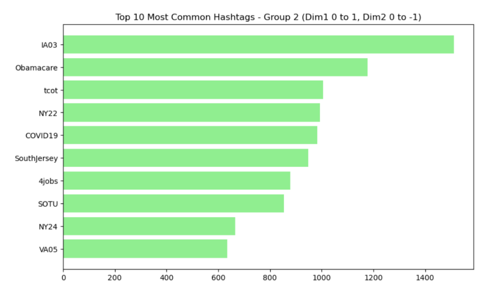

Fig. 4. Top 10 Most Common Hashtags - Group 1 (Dim 1 (0 to 1), Dim 2 (0 to -1)

widespread focus on the virus itself. Political hashtags such as #tcot (Top Conservatives on Twitter), #SOTU (State of the Union), Obamacare, and #ACA (Affordable Care Act) pointed to intense debates surrounding healthcare policies and the pandemic's political implications. Economic anxieties were apparent through 4jobs, as discussions about job loss and economic recovery gained momentum. Broader public health concerns were evident in coronavirus, while regional conversations emerged with hashtags like #utpol (Utah Politics). Lastly, #ForThePeople signaled the rise of grassroots advocacy, pushing for healthcare reform and social justice. This collection of hashtags paints a vivid picture of how deeply the pandemic intertwined political polarization, economic worries, and public health advocacy on social media.

For further analysis the training dataset is divided into four subsets by focusing on the following ideological dimensions:

1) Group 1: Dim 1 (0 to 1), Dim 2 (0 to 1)
2) Group 2: Dim 1 (0 to 1), Dim 2 (0 to -1)
3) Group 3: Dim 1 (0 to -1), Dim 2 (0 to 1)
4) Group 4: Dim 1 (0 to -1), Dim 2) (0 to -1)

Some of the key findings and observations are mentioned below.

- **Political Polarization:** All four groups exhibit a strong emphasis on political issues, indicating that the pandemic has intensified political polarization.
- **Healthcare Focus:** Hashtags like Obamacare, ACA, ProtectOurCare, and Trumpcare are consistently present, underscoring the importance of healthcare policy across all groups.
- **COVID-19 Impact:** The COVID19 hashtag is prominent in all groups, highlighting the pandemic's lasting effect on public discourse.
- **Economic Concerns:** Economic issues, represented by hashtags such as 4jobs, are prevalent across all groups, reflecting significant pandemic-related challenges.
- **Regional and Local Focus:** Groups 2 and 3 emphasize regional or state-level issues, as evidenced by district-specific hashtags.

The findings from this analysis largely align with previous analyses based on individual bar charts, yet specific hashtags and their prominence reveal subtle differences among the groups. Group 1 focuses primarily on national-level political issues and healthcare policy, with less emphasis on regional matters. In contrast, Group 2 emphasizes regional or state-level issues, reflecting a more localized perspective. Group 3 shares this regional focus while also prioritizing healthcare advocacy. Group 4 stands out as more politically active, engaging in discussions about social and environmental issues

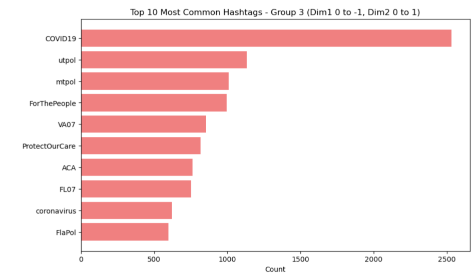

Fig. 5. Top 10 Most Common Hashtags - Group 1 (Dim 1 (0 to -1), Dim 2 (0 to 1)

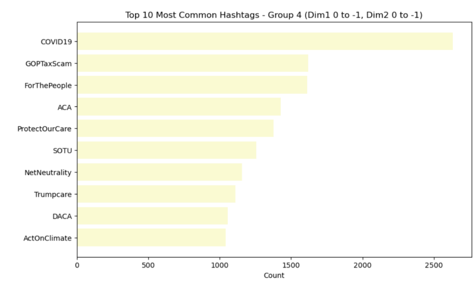

Fig. 6. Top 10 Most Common Hashtags - Group 1 (Dim 1 (0 to -1), Dim 2 (0 to -1)

alongside healthcare and economic concerns.

While all four groups share common themes, the specific focus and emphasis within each group vary. This suggests that the ideological and regional dimensions captured by DW-NOMINATE scores influence the types of conversations and issues discussed within each group. Overall, the analysis highlights the ongoing impact of political polarization, healthcare policy, economic challenges, and regional factors on public discourse related to COVID-19.

In another analysis, the training dataset was divided into two along the 1st dimension. All observations that have a value bigger than zero (in the 1st dimension) was designated as 'conservative', and all other observations were designed as 'liberal'. A ridge plot is then created that shows the changes in these values (1st dimension) through the years for two different ideological groups. The conservative group is identified as red, and the liberal group as blue.

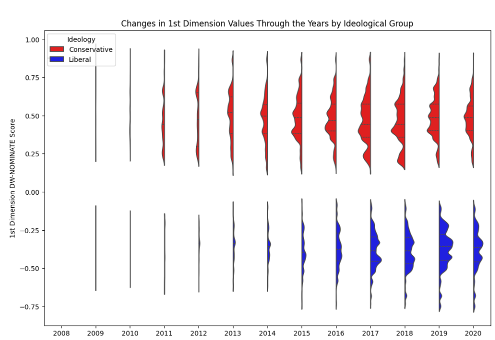

Fig. 7. Changes in 1st Dimension Values Through the Years by Ideological Group

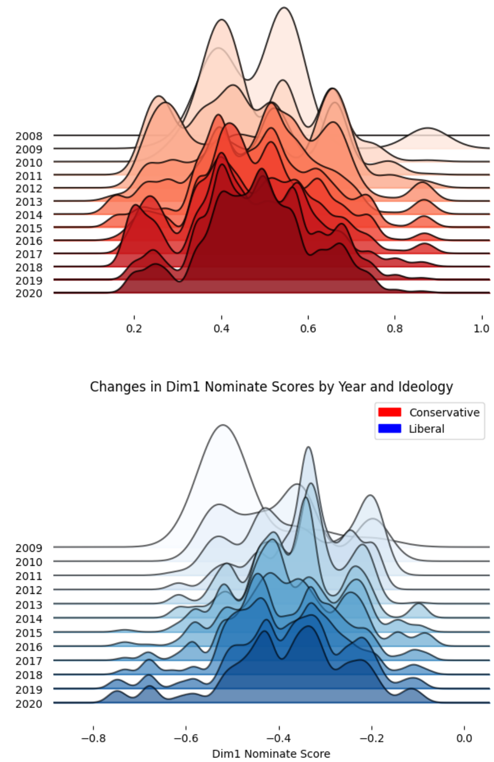

Fig. 8. Changes in Dim1 Nominate Scores by Year and Ideology

The analysis revealed notable shifts in ideological scores over the years for both conservative and liberal groups. For conservatives, there is a significant increase in Dim1 Nominate scores, with earlier years (2008-2010) displaying a broader distribution of lower scores, while more recent years (2016-2020) show a concentration of higher scores, indicating a move towards more extreme conservative positions. In contrast, the Dim1 Nominate scores for liberals remain concentrated around negative values, reflecting consistent liberal positions over time, though there is a slight narrowing of the range in recent years. Additionally, the distribution of conservative scores has become sharper and more skewed towards higher values, suggesting a larger proportion of individuals adopting stronger conservative stances. Conversely, while the liberal distribution remains centered in a similar range, there is a slight increase in score concentration, hinting at a strengthening of liberal positions, albeit not as pronounced as the changes observed among conservatives.

On exploring the data further, we delved into the top-10 tweets that are ideologically most different from each other by calculating the Euclidean distances for the Top-10 most distant tweets along both dimensions, the 1st and the second dimension.

The analysis of the top-10 most distant tweet pairs across both dimensions, the 1st dimension, and the 2nd dimension reveals a diverse ideological landscape within the dataset. Outlier tweets, identified by their frequent appearance in the lists, play a significant role in shaping this landscape. The 1st and 2nd dimensions contribute to the overall ideological

Top 10 Most Distant Tweets (Both Dimensions):
Both Dimensions

	Tweet 1 ID	Tweet 2 ID	Distance
0	109480	16785	1.878528
1	172223	16785	1.878528
2	16785	276367	1.878528
3	109480	126481	1.876241
4	109480	90795	1.876241
5	172223	126481	1.876241
6	172223	90795	1.876241
7	126481	276367	1.876241
8	276367	90795	1.876241
9	109480	198156	1.849152

Fig. 9. Top 10 Most Distant Tweets (Both Dimensions)

Top 10 Most Distant Tweets (1st Dimension):
1st Dimension

	Tweet 1 ID	Tweet 2 ID	Distance
0	109480	126481	1.612877
1	109480	90795	1.612877
2	172223	126481	1.612877
3	172223	90795	1.612877
4	126481	276367	1.612877
5	276367	90795	1.612877
6	109480	16785	1.607965
7	172223	16785	1.607965
8	16785	276367	1.607965
9	134741	126481	1.593604

Fig. 10. Top 10 Most Distant Tweets (1st Dimension)

Top 10 Most Distant Tweets (2nd Dimension):
2nd Dimension

	Tweet 1 ID	Tweet 2 ID	Distance
0	118635	273477	1.654469
1	118635	68452	1.654469
2	118635	24359	1.654469
3	273477	58033	1.654469
4	68452	58033	1.654469
5	58033	24359	1.654469
6	24373	118635	1.648852
7	24373	58033	1.648852
8	118635	132451	1.648852
9	118635	217902	1.648852

Fig. 11. Top 10 Most Distant Tweets (2nd Dimension)

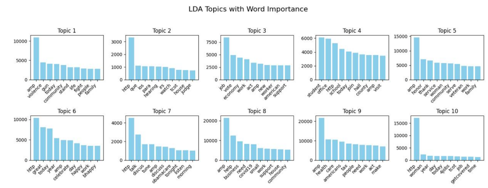

Fig. 12. LDA Topics with word Importance

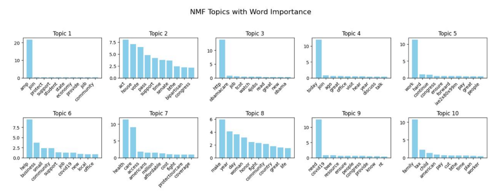

Fig. 13. NMF Topics with word Importance

distance between tweets, suggesting a complex interplay of factors.

Next, we analyse and compare the results of the two models (LDA and NMF) that we have implemented.

LDA Model Results

Latent Dirichlet Allocation (LDA) is a probabilistic generative model that assumes each document is a mixture of various topics. This nature often leads to overlapping words across topics, reflecting how certain terms can relate to multiple themes simultaneously. For example, Topic 1 in the LDA model includes terms like "amp," "trump," "community," "border," and "family," suggesting a blend of social and political discussions, possibly related to government policies. Similarly, Topic 3's focus on words like "health," "care," and "tax" suggests an overlap between healthcare policies and economic discussions. The probabilistic overlap allows words to belong to multiple topics with different probabilities, but it can make topics less distinct. In this model, some terms such as "amp" appear in multiple topics, indicating either their broad applicability or potential noise that could benefit from further preprocessing.

NMF Model Results

Non-negative Matrix Factorization (NMF), in contrast, is a linear algebra-based approach that decomposes the document-term matrix into non-negative matrices. This results in more distinct, non-overlapping topics, as words cannot belong to multiple topics. For example, in NMF, Topic 1 emphasizes terms like "amp," "join," and "community," clearly indicating a focus on social engagement. Likewise, Topic 6, which centers around "health," "care," and "coverage," represents a distinct healthcare-related theme. NMF tends to generate more clearly defined and interpretable topics compared to LDA, with less overlap between terms. This clarity is beneficial when the goal is to separate distinct themes without the complexity of shared terms.

Comparison of LDA and NMF Results

LDA's probabilistic approach allows for more overlap between topics, making it better suited for situations where themes are multifaceted or interconnected, such as in social or political discussions. However, this flexibility comes at the cost of interpretability, as topics may become harder to distinguish due to shared terms. In contrast, NMF produces more distinct topics, making it easier to identify and interpret specific themes. This difference is particularly evident in the healthcare and legislative themes extracted from the dataset, where NMF provides clearer boundaries between topics compared to LDA.

In conclusion, while LDA is valuable for analyzing complex topics with overlapping themes, NMF offers more distinct and interpretable topics, making it a better choice when clear boundaries between themes are important. For this dataset, NMF appears to offer more clarity and less ambiguity in topic extraction.

III. METHOD

For this project, we utilized H2O AutoML to automate the process of model selection, hyperparameter tuning, and training for predicting two target dimensions: dim1_nominate and dim2_nominate. H2O AutoML facilitates the automated exploration of several algorithms, including Gradient Boosting Machines (GBMs), XGBoost, Deep Learning, Generalized Linear Models (GLMs), and Stacked Ensembles. The AutoML process ranks these models based on performance metrics such as RMSE and R^2.

A. H2O AutoML

H2O AutoML was chosen for its efficiency in selecting optimal models based on their performance while minimizing manual intervention. Specifically, it:

- Automatically tuned hyperparameters.
- Considered a broad range of algorithms and compared their performances.
- Created Stacked Ensembles, which combine predictions from multiple base models to achieve superior performance.

The Stacked Ensemble model emerged as the best model for both dimensions, dim1_nominate, and dim2_nominate, as it effectively leveraged the complementary strengths of various algorithms.

B. Strength of the Stacked Ensemble Model

A Stacked Ensemble model aggregates predictions from multiple base learners (e.g., GBMs, XGBoost, GLM) and employs a meta-learner to make final predictions. By combining base learners, the model captures different patterns and complexities in the data, and the meta-learner synthesizes these predictions into a more robust solution. This method helps reduce overfitting and generally improves the model's ability to generalize to unseen data.

In this project, the Stacked Ensembles with GBMs and XGBoost as base models consistently outperformed individual models, demonstrating that combining various algorithms provided a more accurate and generalizable solution.

C. Models Used in H2O AutoML

1) **Gradient Boosting Machines (GBMs)**: GBMs build an ensemble of decision trees sequentially. At each step, a new tree is added to correct the errors made by the previous trees. This can be described as an optimization of a loss function L, where at each iteration t, the goal is to minimize:
$$L = \sum_{i=1}^{n} (y_i - F_t(x_i))^2$$
2) **XGBoost**: XGBoost is an optimized implementation of gradient boosting that incorporates regularization to control overfitting. The objective function it minimizes is:
$$L = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$
where l is a loss function (e.g., squared error), and $\Omega(f_k)$ is a regularization term penalizing model complexity.
3) **Stacked Ensemble**: This model aggregates predictions from multiple base learners, often through a *Generalized Linear Model (GLM)*, which fits a linear combination of the predictions from base models:
$$\hat{y} = \beta_0 + \sum_{k=1}^{K} \beta_k \hat{y}_k$$
where $\hat{y}_k$ are predictions from the base learners, and β_k are coefficients learned by the meta-learner.

IV. RESULTS

A. Dimension 1 (`dim1_nominate`)

Metric	Value
RMSE	0.317
MAE	0.252
R²	0.497

TABLE I

PERFORMANCE METRICS FOR `DIM1_NOMINATE` (TEST SET)

The best-performing model for `dim1_nominate` was a *Stacked Ensemble*, which combined *GBM* and *XGBoost* models. The ensemble achieved an RMSE of 0.346 and an R² value of 0.399 on the cross-validation data.

B. Dimension 2 (`dim2_nominate`)

Metric	Value
RMSE	0.230
MAE	0.177
R²	0.286

TABLE II

PERFORMANCE METRICS FOR `DIM2_NOMINATE` (TEST SET)

For `dim2_nominate`, the *Stacked Ensemble* model achieved an RMSE of 0.238 and an R² value of 0.233 on the cross-validation data.

V. EVALUATION

In the initial approach, the H2O AutoML model discarded the full_text column, which was critical for improving predictive performance. To address this, we utilized a T4 GPU to generate embeddings for the raw text data using the DistilBERT transformer model. The embedding extraction process was handled efficiently using mixed-precision (with torch.cuda.amp) to make full use of the GPU's computational power while minimizing memory overhead.

The embedding process involved encoding the full_text and hashtags columns into numerical vectors using the DistilBERT model, where each text sequence was transformed into a fixed-length embedding of 768 dimensions. This was crucial as these embeddings captured the semantic meaning of the text, which we expected to improve model performance. We processed the text in batches and used techniques such as memory clearing and garbage collection (gc.collect()) to prevent memory overflows during embedding extraction.

Once the embeddings were extracted, they were added as features to both the training and test datasets, replacing the original text columns. The model was retrained using these embeddings as features, but the H2O environment frequently crashed due to memory limitations.

To overcome this, we iteratively reduced the size of the training sample, adjusting the percentage of training data from 100% to 60%. While the model successfully completed training at 60% of the training data, the results were suboptimal. The reduced sample size negatively impacted the model's predictive performance, likely due to a loss of significant information from the text embeddings, which are highly dimensional.

We believe that the likely cause for the underwhelming results can be attributed to the following factors:

- **Dimensionality of Embeddings**: The embeddings generated from *DistilBERT* are 768-dimensional. Reducing the size of the training set may have resulted in an insufficient number of examples to effectively train the model using such high-dimensional data.
- **Memory Constraints**: Despite using the GPU for embedding extraction, the memory limitations of the *H2O* environment posed significant challenges. This forced us to reduce the size of the training dataset, preventing the model from fully utilizing the available features and embeddings.
- **Feature Importance**: The embeddings from `full_text` and `hashtags` may have introduced additional complexity that the model, given the reduced dataset size, was unable to effectively interpret and utilize.

VI. CONCLUSIONS

The use of Stacked Ensemble models, which combine predictions from GBM, XGBoost, and GLM models, yielded the best predictive performance for both dimensions. These models' ability to capture complex patterns in the data, while leveraging the generalization power of ensemble methods, resulted in reduced error metrics and improved accuracy.

Overall, the results indicate that stacking models from different algorithms was an effective approach for this multi-dimensional predictive task.

In the future, we hope to further enhance the model's performance by exploring several strategies. First, we will aim to apply dimensionality reduction techniques such as Principal Component Analysis (PCA) or Truncated SVD to reduce the dimensionality of the text embeddings, which should help alleviate memory constraints while retaining essential information and potentially improving the model's interpretability. Additionally, we hope to focus on increasing resource allocation by expanding the memory resources in the H2O environment or transitioning to a more robust cloud infrastructure, which will allow us to use a larger sample size and improve model stability. Lastly, we hope to explore alternative embedding models, such as BERT, RoBERTa, or GPT-based architectures, which may provide more effective representations of text for our specific task.

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